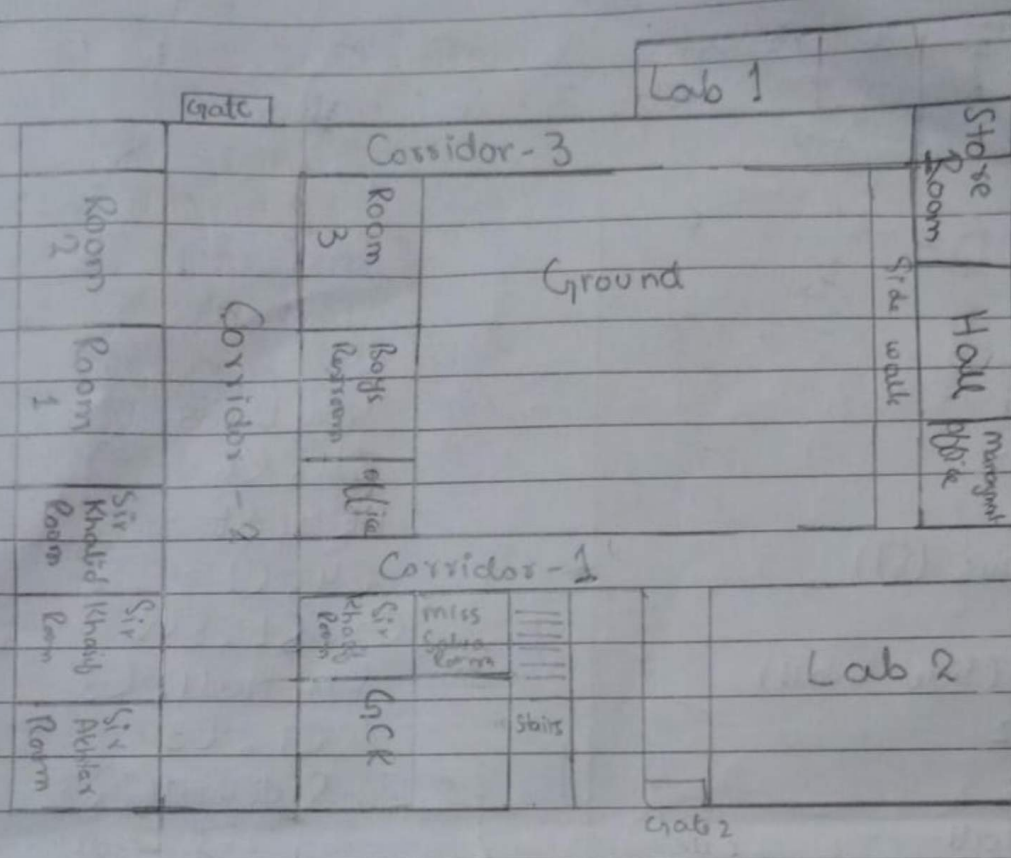
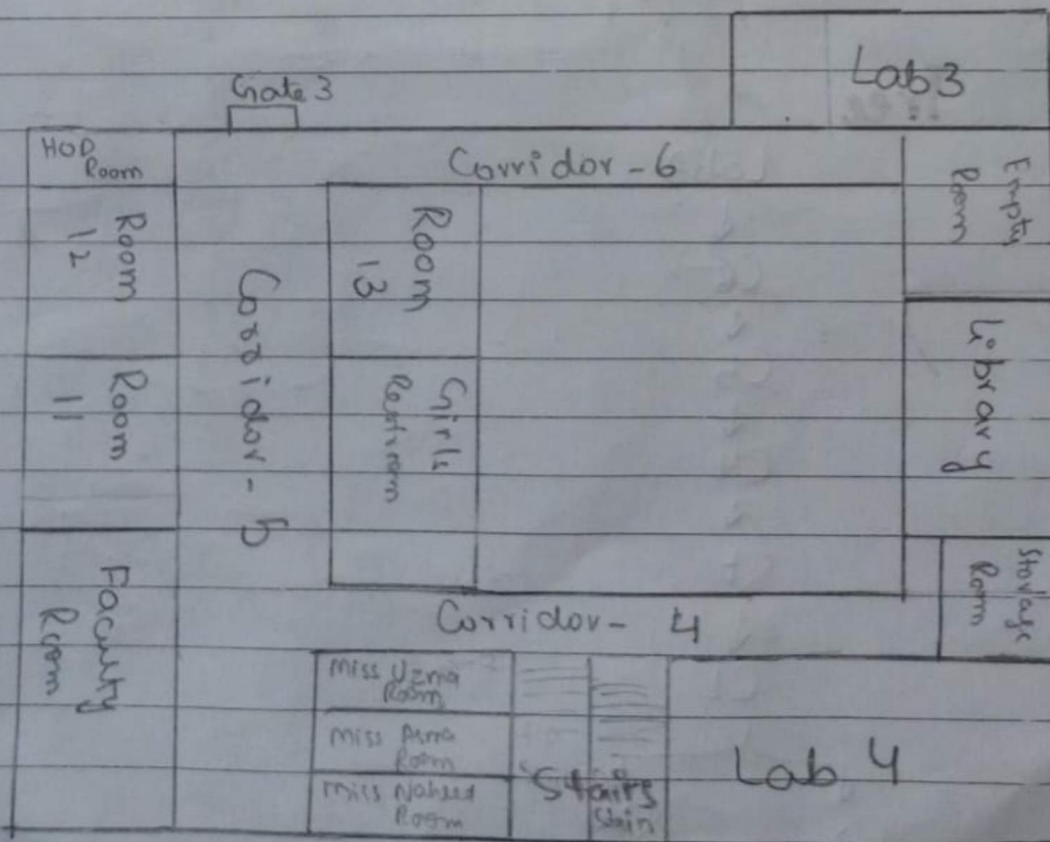


Ground Floor



1st Floor



Graph Representation:

Date _____

Problem

Problem

Lab 3 to Hall

NODE

Connects To

Lab 3

C6 (Corridor 6)

C5

C4

Stairs (ST)

C1

SW (Side Walk)

C2

C3

Hall

C6

C5, Lab3

C4, C6

C5, St

C4, C1

C2, St, SW

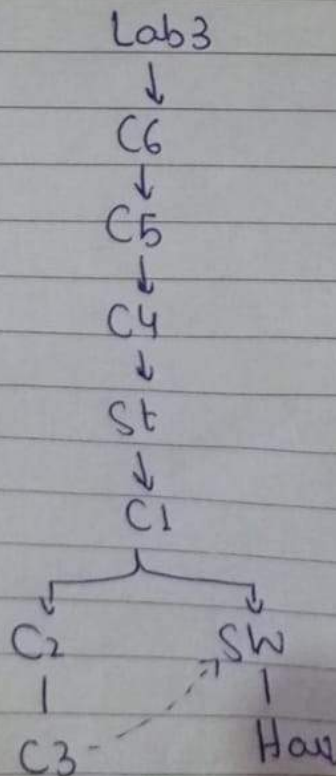
C1, Hall, C3

C1, C3

Side walk, C2

Side walk

BFS Tree



RC

Teacher's Signature _____

BFS Trace:

Step 0 : Initialization:

Q: {Lab 3}

V: {Lab 3}

Dequeue Lab 3:

Q: {C6}

V: {Lab 3, C6}

Dequeue C6:

Q: {C5}

V: {Lab 3, C6, C5}

Dequeue C5

Q: {C4}

V: {Lab 3, C6, C5, C4}

Dequeue C4:

Q: {St}

V: {Lab 3, C6, C5, C4, St}

Dequeue St

Q: {C1}

V: {Lab 3, C6, C5, C4, St, C1}

Dequeue C1

Q: {C2, Sw}

V: {Lab 3, C6, C5, C4, St, C1, C2, Sw}

Dequeue Sw

Q: {C2, Hall, C3}

V: {Lab 3, C6, C5, C4, St, C1, C2, Sw, Hall, C3}

Dequeue C2

Q: {Hall, C3}

V: {Lab 3, C6, C5, C4, St, C1, C2, Sw, Hall, C3}

Dequeue C3:

Q: {Hall}

V: {Lab 3, C6, C5, C4, St, C1, C2, Sw, Hall, C3}

Deque Hall

Q: { Destinations

V: { ----- }

Shortest Path:

Lab 3 $\rightarrow$ C6 $\rightarrow$ C5 $\rightarrow$ C4 $\rightarrow$ St $\rightarrow$ C1 $\rightarrow$ SW $\rightarrow$ Hall